

PATENT DISCLOSURE BRIEF

Presented to Patent Officer · College IP Event · April 16, 2026

APPLICANT	Adarsh Narain Shukla
INVENTION TITLE	Hardware-Entangled Reed-Solomon Erasure Codec with Physical Co-Presence Attestation
FIELD	Cryptography · Coding Theory · Hardware Security · Distributed Storage
FILING STATUS	■ Provisional Application Filed — Priority Date Secured: April 15, 2026
REFERENCE	ANSX-PDB-001

1. THE PROBLEM — WHY THIS INVENTION IS NEEDED

Real-World Incident	Loss	Root Cause
Bybit Cryptocurrency Exchange Hack (2025)	\$1.47 Billion	Private key theft
Ronin Network Breach (2022)	\$625 Million	Private key compromise
Wormhole Bridge Attack (2022)	\$320 Million	Key infrastructure failure
Mt. Gox Exchange Collapse (2014)	\$450 Million	Private key theft
Total documented key-theft losses (2018–2025)	> \$4.2 Billion	No hardware entanglement

In every case above, the attacker obtained a private key and could decrypt all data. **The industry's response for 40 years has been: store keys better.** No system has made the key itself mathematically useless without the physical hardware present. This invention solves that problem for the first time.

2. THE INVENTION — WHAT IS NEW

The Primary Patent Claim (V3.0):

"A data reconstruction apparatus wherein the $GF(q)$ polynomial evaluation coordinates of a Reed-Solomon erasure codec are derived exclusively from a collision-resistant physical co-presence attestation mapping $\Phi : S \rightarrow GF(q)^n$ over a composite hardware state tuple (machine UUID + GPS + NFC HMAC), $|S| \geq 2^{128}$, such that any incorrect hardware state produces a non-zero algebraic residual in the overdetermined Vandermonde decoding system — constituting a mathematical proof of hardware-state inconsistency."

In Plain English:

The file is sharded across 12 cloud locations. The mathematical parameters that allow reconstruction are derived from three physical sources: **this specific Mac + this specific NFC sticker + this specific GPS location**. If even one of these is wrong or missing, the mathematics of reconstruction provably fails — not because of a password, not because of an access control rule, but because the data's algebraic geometry requires the exact hardware to exist.

Key Novelties:

- **First in 65 years of Reed-Solomon literature** to use physical hardware telemetry as GF evaluation coordinates
- **Ghost Map (Steganographic Manifest):** One innocuous image secretly contains the master key + 12 cloud shard addresses
- **Blockchain PKI:** Ethereum Smart Contract maps usernames to RSA-4096 public keys — trustless, serverless identity

■ **Zero-Step NFC Provisioning:** iPhone scans QR → downloads generated Shortcut → writes NFC sticker automatically

3. PRIOR ART SEARCH — RESULTS

A comprehensive search was conducted across IEEE Xplore, arXiv (cs.CR, cs.IT), ACM Digital Library, USPTO, Google Patents, and Espacenet. **14 references were identified and analyzed.** None anticipates the primary claim.

Reference	Closest To	Anticipates?	Why Distinct
Reed & Solomon (1960)	L1 — RS codec	■ NO	Fixed eval. coords; no hardware
Suh & Devadas (2007) — PUF + BCH	L2 — hardware + ECC	■ NO	ECC stabilizes hardware; claimed: hardware defines ECC
US8887307B2 — GPS + UUID auth	L3 — multi-modal tuple	■ NO	Produces password, not GF coordinates
US8713329B2 — GF(q) secret share	L1 + L5 structure	■ NO	Fixed random eval. points; no hardware connection
US11037227B1 — Storj (RS + blockchain)	L1 + Claim 3	■ NO	Blockchain stores keys; not evaluation coordinates
US20150006390A1 — Multi-item stego	Claim 2 — stego	■ NO	Payment credentials; not cloud shard manifest

4. NON-OBVIOUSNESS — SECTION 2(1)(ja)

Argument	Summary
Synergistic Result	RS literature has used FIXED evaluation coordinates for 65 years. Using hardware telemetry produces a wholly new p
Teaching Away	RS literature explicitly requires fixed public evaluation points. NFC standards explicitly separate hardware authentication
No Motivation to Combine	The invention requires simultaneous expertise across 5 unrelated disciplines: coding theory, cryptography, steganogra
Unrecognised Problem	'Make a private key mathematically useless without hardware' was never formulated as a design goal in 50 years of co
Long-Felt Need	Private key theft has caused \$4.2B+ in documented losses. For 40 years, the industry's only response was better key

5. DOCUMENTS AVAILABLE FOR INSPECTION

#	Document	Content	Status
1	ANSX-NOB-001 Non-Obviousness Declaration	8-part legal + technical argument against all §103 rejections	■ Complete
2	ANSX-PAR-001 Prior Art Search Report	14 references, element-by-element L1–L5 claim chart analysis	■ Complete
3	patent_claims.md Provisional Specification	Primary + 7 dependent claims, technical description, drawings ref.	■ Complete
4	Blockchain Timestamp OrigineStamp Certificate	SHA-256 hash of all documents anchored to Bitcoin blockchain	■ Complete

5	IPO Provisional Receipt	Official priority date: April 15, 2026	■ Filed Today
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■ NOVEL · INVENTIVE · INDUSTRIALLY APPLICABLE — RECOMMEND GRANT

Applicant Declaration	All information in this disclosure is accurate and complete to the best of my knowledge. This invention has been rec
Signature:	Adarsh Narain Shukla
Date:	April 15, 2026
Contact:	[available on request]

This document is a Provisional Patent Disclosure Brief prepared under the Indian Patents Act, 1970. Full patent application to be filed within 12 months of provisional filing date.